

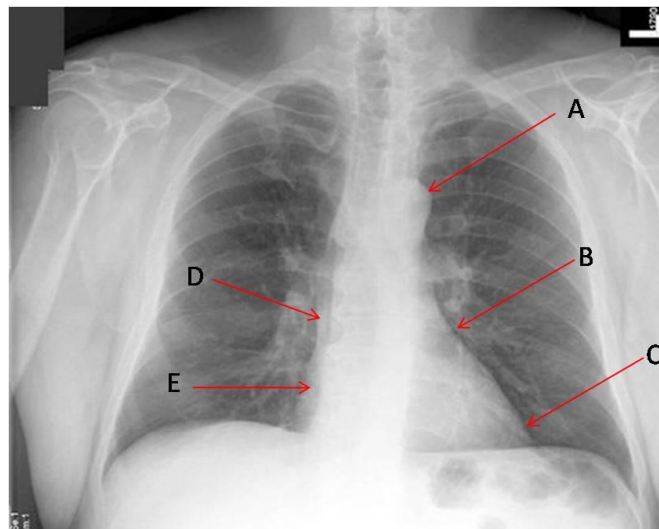
CPR Esoft Quiz 1

Question #: 1

A 50-year-old man is brought to the emergency department by the paramedics because of difficulty breathing immediately after being involved in a motor vehicle collision. His blood pressure is 70/50 mmHg, pulse is 105/min and respirations are 22/min. Venous return to the heart is blocked. In the attached radiograph shown, which of the following areas is most likely affected?

- A. A
- B. B
- C. C
- D. D
- E. E

Attachment:



chest x ray.jpg

Question #: 2

A newborn boy is examined by a physician. Auscultation of his chest shows a holosystolic heart murmur. Ultrasonography shows an open structure connecting the aorta and the pulmonary trunk. Which of the following is the most likely embryological origin of this structure?

- A. Left dorsal aorta
 - B. Right dorsal aorta
 - C. Left fifth aortic arch
 - D. Left sixth aortic arch
 - E. Right sixth aortic arch
-

Question #: 3

A premature newborn boy is admitted to the neonatal intensive unit because of severe respiratory distress. Despite medical intervention, He dies within 24 hours. The postmortem examination shows the cause of death is respiratory distress syndrome. There is an incidental finding of an atrial septal defect (ASD) that has little to no hemodynamic/clinical significance. Which of the following types of ASD is most likely?

- A. Ostium secundum
 - B. Probe patent oval foramen
 - C. Common atrium
 - D. Endocardial cushion with ostium primum
 - E. Sinus venosus
-

Question #: 4

A 59-year-old woman comes to the physician for a routine examination. Physical examination shows a systolic murmur which is loudest at the second intercostal space on the right. Echocardiography shows valvular stenosis. Which of the following chambers or vessels most likely receives the blood that is being ejected through the narrowed opening?

- A. Left ventricle
 - B. Right ventricle
 - C. Aorta
 - D. Pulmonary trunk
 - E. Left atrium
-

Question #: 5

A 75-year-old woman is brought to the emergency department due to a feeling of pressure-like chest pain beneath the sternum, which radiates to her left arm. An ECG indicates a myocardial infarction, along with a right bundle branch block. Physical examination shows a pronounced second heart sound. Which pair of heart valves' closure is primarily responsible for generating this second heart sound?

- A. Aortic and pulmonary
 - B. Aortic and tricuspid
 - C. Tricuspid and mitral
 - D. Mitral and pulmonary
 - E. Tricuspid and pulmonary
-

Question #: 6

A 14-year-old boy is brought to the physician for a routine examination. Physical examination shows a regurgitation murmur of his bicuspid valve. Into which of the following chambers or vessels is the blood most likely regurgitating?

- A. Left atrium
 - B. Right atrium
 - C. Pulmonary trunk
 - D. Aorta
 - E. Left ventricle
-

Question #: 7

A 48-year-old man comes to the physician because of a 4-month history of shortness of breath on exertion. Coronary angiography shows severe occlusion of the anterior interventricular artery. He is taken for coronary bypass surgery. Which of the following cardiac veins most likely accompanies the blocked artery?

- A. Middle
- B. Anterior
- C. Posterior
- D. Coronary sinus
- E. Great

Question #: 8

A 56-year-old man is admitted to the hospital for surgery to correct a congenital atrial septal defect. During the surgery, the surgeon must be careful to stay away from the blood supply to the conducting system of the heart including the supply to the SA and AV nodes. Which of the following vessels should most likely be avoided?

- A. Anterior interventricular
- B. Left coronary
- C. Right coronary
- D. Circumflex
- E. Posterior interventricular

Question #: 9

A 50-year-old man is brought to the emergency department by his wife because of a 3-hour history of substernal chest pain that radiates into his neck and left upper limb. ECG shows ST elevation in the precordial leads. A diagnosis of myocardial infarction is made. Which of the following structures most likely carries the visceral afferent fibers responsible for the patient's pain?

- A. Intercostobrachial nerve
- B. Cardiopulmonary nerves
- C. Greater splanchnic nerves
- D. Phrenic nerve
- E. Grey rami communicans

Question #: 10

A 2-day-old girl is admitted to the cardiology unit because of a 6-hour history of labored breathing and cyanosis. Echocardiography was performed, which shows a congenital heart defect that results in a right-to-left shunt. Which of the following conditions is most likely to produce this type of shunt at birth?

- A. Tetralogy of Fallot
- B. Atrial septal defect

- C. Ventricular septal defect
 - D. Patent ductus arteriosus
 - E. Patent foramen ovale
-

Question #: 11

A 50-year-old man is brought to the emergency department because of a 3-hour history of substernal chest pain that radiates into his neck and left upper limb. ECG shows ST elevation in the precordial leads. A diagnosis of myocardial infarction is made. Which of the following structures is most likely responsible for the pain being referred to the upper limb?

- A. Intercostobrachial nerve
 - B. Cardiopulmonary nerves
 - C. Greater splanchnic nerves
 - D. Phrenic nerve
 - E. Grey rami communicans
-

Question #: 12

A 2-year-old boy is brought to his physician by his mother because of a cold. His mother says that he has had frequent episodes of "the cold" since birth. Physical examination shows decreased breath sounds in both lungs and heart sounds heard best on the right chest. Chest x-ray shows the apex at the right 4th intercostal space. Defect in which of the following processes resulted in his findings?

- A. Lateral folding
 - B. Head to tail folding
 - C. Formation of pleuropericardial folds
 - D. Dextro-looping
 - E. Canalization of primordial heart tube
-

Question #: 13

A 58-year-old man comes to the physician because of a 2-month history of difficulty swallowing. Imaging studies show an enlargement of the most posteriorly located chamber of heart which is impinging on the esophagus resulting in his symptoms. Which of the following parts of the heart

is most likely in a position to cause this impingement?

- A. Left atrium
 - B. Right atrium
 - C. Left ventricle
 - D. Right ventricle
 - E. Aortic arch
-

Question #: 14

A 55-year-old man comes to the physician for a follow-up examination. He performs poorly on an exercise stress test. Coronary angiography shows significant blockage of one of the major arteries supplying his heart. He is sent for a coronary artery bypass procedure. During the surgery, after the chest is opened using a midline sternotomy (splitting of the sternum), the pericardial sac is opened by a longitudinal incision. If a horizontal incision has been used on the pericardium, which of the following structures would be most likely be at risk?

- A. Vagus nerve
 - B. Internal thoracic artery
 - C. Azygos vein
 - D. Thoracic duct
 - E. Phrenic nerve
-

Question #: 15

A 78-year-old female presented with edema of the left upper limb due to poor venous return. Examination revealed an aneurysm of the ascending aorta that was impinging on a large vein lying immediately antero-superior to it. Which of the following veins is most likely being compressed?

- A. Azygos
 - B. Internal thoracic
 - C. Left brachiocephalic
 - D. Left superior intercostal
 - E. Right brachiocephalic
-

Question #: 16

A 56-year-old man is brought to the emergency department by his son because of a 2-day history of difficulty breathing. Two months ago, he was diagnosed with a mediastinal tumor. Imaging studies show that cancer has invaded the nerve that travels anterior to the hilum of the lung. Which of the following nerves is most likely affected?

- A. Vagus
 - B. Cardiopulmonary
 - C. Phrenic
 - D. Greater thoracic splanchnic
 - E. Intercostobrachial
-

Question #: 17

A 2-month-old infant is brought to the physician due to poor feeding and rapid breathing. Physical examination shows a systolic murmur and cyanosis. Echocardiography shows a large ventricular septal defect (VSD) and an overriding aorta. Given these findings, which of the following embryological events is most likely responsible for this infant's condition?

- A. Abnormal development of endocardial cushions
 - B. Incorrect division of the truncus arteriosus
 - C. Failure of the foramen ovale to close after birth
 - D. Abnormal development of the sinus venosus
 - E. Insufficient formation of the bulbus cordis
-

Question #: 18

A 45-year-old woman comes to the physician because of a 2-week history of worsening left sided upper limb edema. She has a history of left breast cancer for which she received surgical treatment including left axilla lymph node dissection. Physical examination show 1+ pitting edema in the left upper extremity with no erythema or increased warmth. Which of the following vessels is most likely affected in this case?

- A. Lymphatic capillaries
- B. Fenestrated capillaries
- C. Sinusoidal capillaries

- D. Elastic artery
 - E. Large vein
-

Question #: 19

A 64-year-old woman is brought to the emergency department because of a sudden onset of chest pain, nausea, vomiting, and blurred vision. She has a history of essential hypertension and is non-adherent to her medication. Her blood pressure is 180/120 mm Hg, and pulse is 92/min. Physical examination shows crackling sounds in both lung fields. She is diagnosed with hypertensive emergency and started on hydralazine (an antihypertensive medication). This medication is known to regulate vascular tone by causing widening of blood vessels (vasodilation) by acting directly on the smooth muscle of the vessel wall. Which of the following best identifies the layer of the vessel wall in which this muscle type predominates?

- A. Media
 - B. Epicardium
 - C. Adventitia
 - D. Intima
 - E. Endocardium
-

Question #: 20

A 78-year-old woman comes to the physician because of a 2-week history of brief fainting episodes while climbing steps and occasional lightheadedness during prolonged walking, and a 3-month history of worsening shortness of breath during regular house cleaning activities. She was diagnosed with angina pectoris (chest pain brought on by exertion and relieved with rest) 3 months ago. Physical examination shows bilateral swollen ankles and an irregular heartbeat. Cardiac imaging studies show a "floppy" aortic valve. Examination of a biopsy specimen obtained from the affected valve show pathological accumulations of proteoglycans with degradation of collagen and elastic fibers. Which of the following best characterizes the layer of the heart valve in which the accumulated extracellular matrix component is normally found?

- A. Simple projections of the tunica intima
 - B. Lined by endothelium
 - C. Lacks collagen fibers
 - D. Lacks elastic fibers
 - E. Functions as a shock absorber
-

Question #: 21

A 42-year-old man is brought to the emergency department because of a 1-hour history of seizures, and a 2-day history of headaches and disorientation. He has a history of a viral illness 2 weeks ago. His temperature is 104°F (40°C), and pulse is 102/min. Physical examination shows an unconscious, febrile man. He dies despite appropriate medical care. Histopathological examination of biopsy specimens of the brain shows necrotic changes to the endothelial cells that line the lumen of the vessel that forms part of the blood-brain barrier. Which of the following best identifies an alternative location for the most likely affected type of blood vessel?

- A. Ciliary processes of the eye
 - B. Glomeruli of the kidneys
 - C. Spleen
 - D. Liver
 - E. Muscle
-

Question #: 22

A 62-year-old man is brought to the emergency department because of a 2-day history of fevers, chills, and generalized weakness. Physical examination shows a weak man that is warm to the touch. Laboratory studies show an elevation in both the white blood cell count, and biomarkers for endothelial activation such as von Willebrand factor. Which of the following is most likely a normal function of the activated tissue?

- A. Production of platelets for blood clotting
 - B. Secretion of vasoactive substances such as nitric oxide
 - C. Production of leukocytes
 - D. Production and secretion of erythrocytes
 - E. Maintains a pro-thrombogenic surface at all times
-

Question #: 23

A 52-year-old man comes to the physician for a follow up examination. He has a 15-year history of high blood pressure. His pulse is 82/min, blood pressure is 142/98 mm Hg, and respirations are 12/min. Physical examination shows a split S2 sound. Which of the following cardiac events is most likely associated with this heart sound during a cardiac cycle?

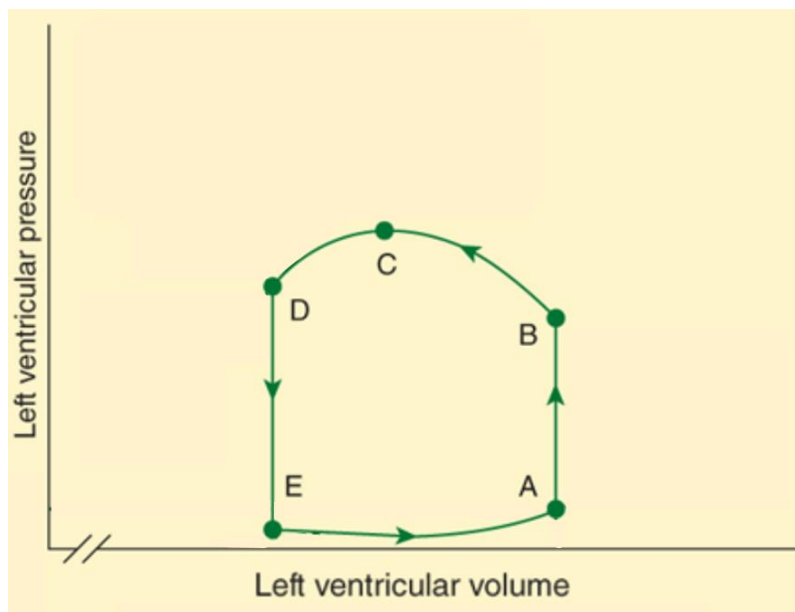
- A. Closure of the mitral valve
- B. Opening of the tricuspid valve
- C. Closure of the aortic and pulmonic valves
- D. Inrushing blood into the ventricle during atrial systole
- E. Rapid ejection of blood into the aorta

Question #: 24

A 35-year-old man comes to the physician because of a 2-month history of shortness of breath. He has had rheumatic fever in his childhood. His vital signs are within normal limits. Physical examination shows an opening snap followed by a rumbling mid-diastolic murmur in the mitral area. Echocardiography shows a stenotic mitral valve. Which of the following labelled points best represents the opening of the affected valve in the attached pressure-volume loop figure?

- A. A
- B. B
- C. C
- D. D
- E. E

Attachment:



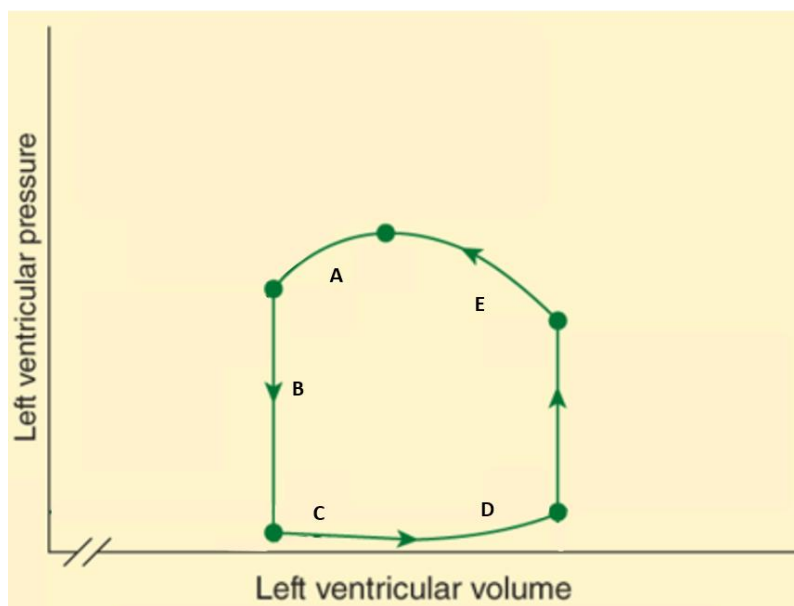
Capture 2.PNG

Question #: 25

A 65-year-old woman is brought to the emergency department because of a 2-hour history of chest pain and shortness of breath. She has a 15-year history of hypertension and type 2 diabetes mellitus. The pain started when the patient was walking up the hill. It was in the anterior chest wall with radiation to the left upper limb and the left jaw. Her pulse is 86/min, blood pressure is 148/94 mmHg, and respirations are 18/min. Physical examination shows crepitations in the basal lungs and a loud S3 in the mitral area. The pressure-volume loop of the patient is shown in the attached figure. Which of the labeled parts of the graph is most likely related to the abnormal heart sound heard in this patient?

- A. A
- B. B
- C. C
- D. D
- E. E

Attachment:



Picture3.png

Question #: 26

A 62-year-old man is brought to the intensive care unit because of overwhelming hypotension and fever. The patient has received 4 L of normal saline bolus fluids. Empirical antibiotics were begun with no improvement in his condition. His blood pressure is 60/30 mm Hg, pulse is 112/min, temperature is 40.6° C (105° F), and respiratory rate is 23/min. The physician orders intravenous norepinephrine. Which of the following is a direct therapeutic effect of norepinephrine in this clinical scenario?

- A. Bradycardia

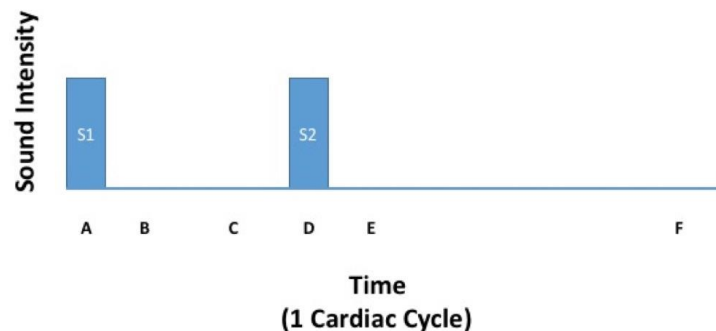
- B. Bronchoconstriction
- C. Coronary vasodilation
- D. Decreased inotropy
- E. Peripheral vasodilation

Question #: 27

A 59-year-old man comes to the physician because of 3-week history of breathlessness and severe tiredness on exertion. His pulse is 90/min and blood pressure is 130/50 mm Hg. Physical examination shows a murmur on auscultation that is made worse during maneuvers to increase peripheral resistance, such as squatting and hand grip exercises. Echocardiography shows an increased left ventricular end diastolic volume. Using the attached diagram of a cardiac cycle cardiophony, during which of the following phases is the murmur most likely to be heard loudest?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F

Attachment:



cardiac cycle cardiophony.jpg

Question #: 28

A 49-year-old man comes to the physician because of a 2-week history of episodic dizziness and palpitations. He says the duration varies from seconds to a few minutes. He is given a continuous monitoring device to wear for 48 hours. The post-monitoring report shows a pulse of 145/minute and blood pressure of 95/75 mm Hg, during an episode of dizziness. Using the attached chart, which of the following changes in cardiovascular parameters is most likely during

one of these episodes?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F
- G. G
- H. H

Attachment:

	Mean Arterial Pressure	Pulse Pressure	Stroke Volume	Baroreceptor Firing Rate
A	No change	↓	↓	↓
B	↓	No change	↓	↓
C	↓	↓	No change	↓
D	↓	↓	↓	↓
E	↑	↑	↑	↑
F	No change	↑	↑	↑
G	↑	No change	↑	↑
H	↑	↑	No change	↑

cv table.jpg

Question #: 29

A 61-year-old man comes to the physician because of a 6-month history of progressively worsening shortness of breath. Physical examination shows a late systolic crescendo murmur with a midsystolic click on auscultation of the heart. The murmur is best heard over the apex, is loudest at S1, is shortened with squatting, and is longer and more intense when venous return is decreased by standing or a Valsalva maneuver. Which of the following is the best index of the preload on this patient's heart?

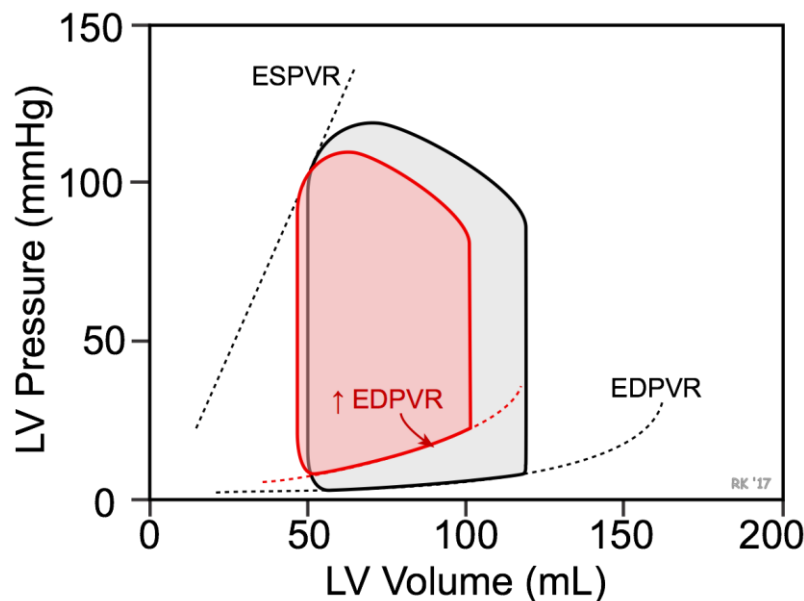
- A. Mean arterial pressure
- B. Central venous pressure
- C. Pulmonary capillary wedge pressure
- D. Ventricular end-diastolic pressure
- E. Ventricular end-diastolic volume

Question #: 30

A 55-year-old man is brought to the emergency department by his wife because of a 4-hour history of shortness of breath and confusion. He was diagnosed with heart failure due to diastolic dysfunction 3 years ago. Physical examination shows rales in the basal lung fields bilaterally. A pressure-volume loop (red) of the patient is shown in comparison with a normal loop (black) in the attached figure. Which of the following values best estimates ejection fraction of this patient?

- A. 20%
- B. 30%
- C. 40%
- D. 50%
- E. 60%

Attachment:



hf006-pv-loop-diastolic-dysfx.png